Construct-Specified LLM Measurement for Text Artefacts

From holistic prediction to validity-guided measurement

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Second-reader overview

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Motivation

The challenge of operational text evaluation

- LLMs increasingly produce open-ended text in operational settings.
- In these settings, evaluation must support decisions about individual outputs: accept, revise, rank, diagnose, or improve.
- A single holistic score can rank outputs, but does not explain *why* an artefact is good or bad.
- To diagnose quality, "good" must be decomposed into dimensions that constitute the target construct.
- **Problem**: without explicit construct specification, multi-dimensional scores are difficult to interpret as meaningful measurements.

Thesis motivation

Artefact-level evaluation requires explicit construct specification: defining quality dimensions, measuring them, and validating whether they support the intended quality claim.

Literature lineage

Why artefact-level LLM measurement matters

- Classic evaluation methods were limited to surface-level overlap [1, 2] or semantic similarity [3, 4]
- LLMs addressed these limitations by acting as **general-purpose evaluators** [5, 6]
- Current frontier: automatically evaluating text on multiple dimensions [7–9]
    - This adheres to the multi-dimensional nature of text quality [10]

Core problem

Multi-dimensional LLM evaluation produces criterion-level scores, but it is often unclear how these scores should be recomposed into a meaningful claim about artefact quality.

Thesis response

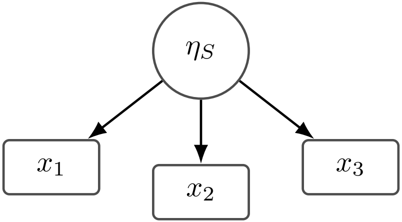
Define artefact evaluation as a measurement problem: quality dimensions are theoretically specified, measured separately with LLMs, and recomposed to collect validity evidence for the resulting construct-level measurement claim.

The measurement problem: three requirements

1. **Target construct**: what is being measured?
2. **Measurement model**: how do scores relate to the construct?
3. **Validity evidence**: what evidence supports the interpretation?

- Evaluation can target **systems** or **artefacts**: model capability across tasks vs. quality of an individual text.
- Psychometrics distinguishes between **reflective** and **formative** measurement models [11].
- These models define how scores relate to the construct, what the main validity risks are, and what kind of evidence is needed.

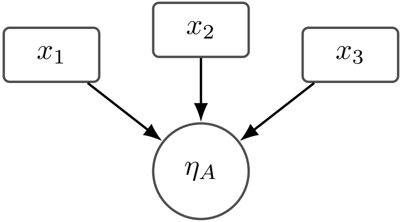
System-level evaluation
Reflective measurement



Latent system ability η_S causes observed performance across tasks, prompts, or benchmark items.

$$x_j = \lambda_j \eta_S + \varepsilon_j$$

Artefact-level evaluation
Formative measurement



Criterion-specific scores jointly constitute the quality η_A of an individual artefact.

$$\eta_A = \sum_{j=1}^{J} \gamma_j x_j + \zeta$$

Measurement framework

Implication for the empirical design

- In formative measurement, the criterion set is part of the construct specification: changing the criteria changes the construct [11]
- Validity evidence is collected by relating LLM-derived measurements to external validation criteria
- The empirical question is whether explicit formative specification yields stronger validity evidence and more interpretable diagnostics.

Research questions
Evaluating construct specification

RQ1 Does explicit formative recomposition of theoretically specified criteria provide stronger validity evidence than holistic or weakly specified alternatives?

RQ2 Do separately elicited formative criteria support more diagnostically interpretable artefact-level measurements than holistic or weakly specified alternatives?

Measurement procedure

From construct specification to validity evidence

1. Theoretical Specification

Construct specification

Define the target construct and the criterion set

2. LLM Measurement

LLM scoring & derivation

Score each criterion with the LLM

Derive expected values & entropy

3. Empirical Evaluation

Evaluation of measurements

Assess validity evidence and diagnostic interpretability

Key idea

The procedure turns a target construct into criterion-level LLM measurements, then evaluates whether recomposed scores provide validity evidence and diagnostic interpretability.

Validation Settings & Experimental Conditions

Testing LLM measurements against external benchmarks

FeedbackQA

Public Q&A dataset

Benchmark: **human ratings**

Specifications compared:

- Naive holistic (1 score)
- Informed holistic (1 score)
- Formative criteria (4 scores)

Barter Deals

Proprietary marketplace data

Benchmark: **structural market baseline**
Outcome: application counts

Specifications compared:

- Naive holistic (1 score)
- Macro-formative (3 scores)
- Fine-grained formative (10 scores)

Key comparison

Do explicit formative LLM measurements provide stronger validity evidence than weaker construct specifications or non-LLM baselines?

Main results

Fine-grained measurement helps most when the construct is separable

FeedbackQA

Convergent validity

- Formative $>$ holistic baselines
- Best Spearman's ρ: **0.578**
  - Human-human ρ: 0.541
- Criteria are strongly intercorrelated

Interpretation:
Decomposition improves alignment with human ratings, but dimensions are not cleanly separable.

Barter Deals

Predictive validity

- Fine-grained formative scores add strong incremental fit
- Best ΔBIC: **+352**
- Poisson deviance: **14.68** $\rightarrow$ **13.23**

Interpretation:
Fine-grained criteria capture meaningful variation in creator application behavior and are cleanly separable.

Conclusion

Moving from holistic prediction to meaningful measurement

The Core Takeaway

Formative LLM measurement provides stronger validity evidence than holistic baselines, but its utility is construct-dependent: it works best when the dimensions are empirically separable.

Broader Implications for the Field:

- **Epistemic shift:** LLM evaluation should be treated as formal measurement, not merely text prediction.
- **Validity framing:** Human ratings and behavioral outcomes provide validity evidence, not infallible "true scores."
- **Practical design:** Explicit, fine-grained criteria improve both statistical validity and diagnostic transparency.

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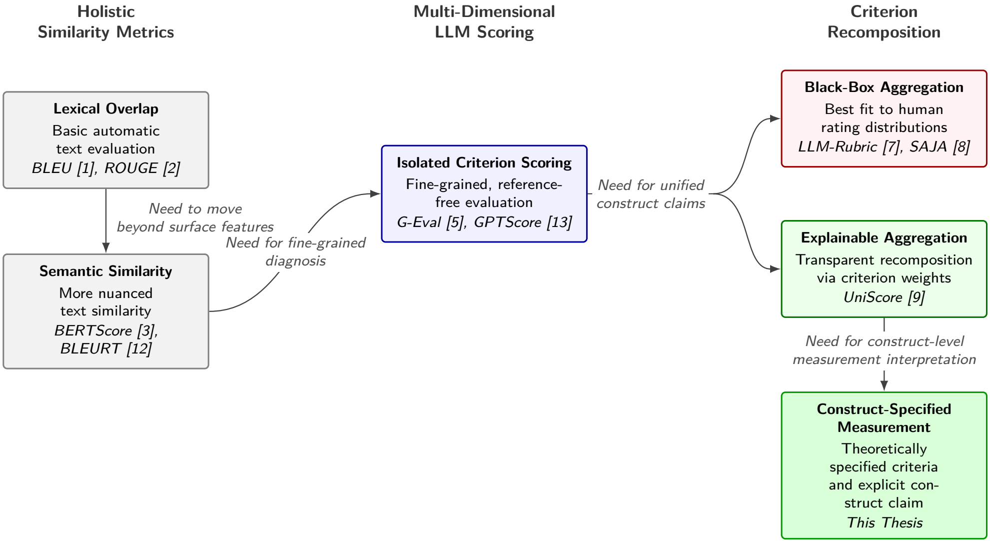
Appendix: Literature lineage



Appendix: Full measurement framework